

Prit Patel

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Professional Summary

Customer-obsessed Forward Deployed Engineer and Computer Science student (GPA: 9.67/10) who combines deep full-stack and **Generative AI engineering expertise** with a strong drive to solve real customer problems hands-on. Proven ability to rapidly prototype, deploy, and iterate on complex AI-powered systems — from LLM pipelines and RAG architectures to real-time WebSocket platforms and autonomous agent workflows. Comfortable moving fluidly between **scoping customer requirements, architecting technical solutions, writing production code, and explaining complex systems** to both technical and non-technical stakeholders. Solved **400+ LeetCode problems**, reflecting the kind of structured, first-principles thinking that FDEs rely on under pressure.

Core Competencies

- **Customer Engagement:** Requirements gathering, technical scoping, solution demos, stakeholder communication (technical & non-technical audiences)
- **Rapid Prototyping & Deployment:** End-to-end feature delivery from zero to production; ability to build, test, and ship working demos under tight timelines
- **Full-Stack Engineering:** Next.js, React.js, Node.js, Express.js, FastAPI (Python), TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS
- **AI & LLM Integration:** Groq (LLaMA 3.3), Gemini 1.5 Pro, OpenAI API, LangChain, RAG Pipelines, Qdrant, NVIDIA NIM, Prompt Engineering, Multi-Agent Systems
- **Backend & APIs:** REST APIs, WebSockets, SSE Streaming, WebRTC, real-time system design
- **Databases:** PostgreSQL (Supabase), MongoDB, MySQL; schema design, RLS, query optimization
- **Systems Thinking:** Architecture design, debugging complex distributed systems, root-cause analysis
- **Tools & Platforms:** Git, GitHub, Docker (Basics), Vercel, Supabase, Stripe, AWS (Basics)
- **Languages:** TypeScript, JavaScript (ES6+), Python, C++

Projects

Kyren — AI-Powered Learning Platform (End-to-End Product Build) 2026 – Present

- **Scoped and architected** a multi-user SaaS platform from scratch for four distinct user personas (students, teachers, tutors, admins) — translating complex, multi-stakeholder requirements into a clean, role-isolated technical architecture using Next.js, TypeScript, and Supabase PostgreSQL
- **Shipped a production LLM pipeline** (planning → lesson generation → enrichment → quiz creation) that generates complete multi-module courses from any topic in **under 60 seconds**, turning a vague user need ("I want a course on X") into a fully automated, deployable solution
- **Solved a real accessibility problem** by building a multimodal AI doubt assistant that accepts handwritten note images via OCR and Gemini 1.5 Pro vision — removing barriers for students without reliable internet or typed-text access
- **Delivered an in-browser AI proctoring system (ProctorX)** using TensorFlow.js and MediaPipe, replacing the need for external proctoring software — reducing cost and deployment complexity while achieving real-time 30 fps inference
- **Integrated end-to-end Stripe billing** with webhook-driven subscription lifecycle management, enabling the platform to scale commercially across individual and institutional tiers

CodeGuard — AI Code Security Platform (Customer Problem → Shipped Solution) Dec 2025 – Present

- **Identified a high-impact developer pain point** — insecure code reaching production due to rushed or shallow PR reviews — and independently designed and built an LLM-powered solution to address it end-to-end
- Engineered a **real-time PR analysis pipeline** that ingests pull request diffs, applies LLM reasoning to detect security vulnerabilities and logic errors, and surfaces actionable risk reports before merge — fitting naturally into existing CI/CD workflows without disrupting developer experience

- Demonstrated the **FDE instinct** of owning both the customer problem and the technical solution: scoped the use case, chose the right architecture, built and deployed the product independently

ARIA v2.0 — Autonomous Agentic Research Platform (Complex System Delivery) 2026 – Present

- **Architected and shipped** a production multi-agent research system where a master LLM decomposes complex queries into parallel worker agent subtasks — synthesizing large-scale web intelligence across hundreds of sources
- **Built Ghost Executive**, an autonomous agent that takes a customer's resume PDF as input and independently handles job discovery, profile parsing, and application form automation — a full workflow automation from a single user input
- **Debugged and resolved** cross-environment reliability issues in the browser automation backend (stealth profiles, SPA navigation, infinite scroll, CAPTCHA resilience), ensuring consistent delivery across dynamic and protected web targets
- Selected **Groq (LLaMA 3.3) for latency-critical planning** and **Gemini 1.5 Pro for vision/OCR tasks** — demonstrating the model-selection judgment that FDEs apply when matching tools to customer performance requirements

StockMind — Multi-Agent Market Simulation (High-Complexity Technical Delivery) 2026 – Present

- **Designed and delivered** a highly complex concurrent simulation system using FastAPI and WebSockets — orchestrating thousands of LLM-driven agent decisions per tick and streaming live market state to clients with sub-100ms latency
- Built a **real-time RAG + sentiment pipeline** (NVIDIA NIM embeddings + Groq LLM) that ingests live news, retrieves relevant historical context, and produces structured sentiment signals — demonstrating ability to integrate multiple AI services into a single coherent customer-facing feature

Education

B.Tech in Computer Science and Engineering 2023 – 2027

CHARUSAT University, Anand, Gujarat

GPA: 9.67 / 10.0 · Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering, Machine Learning

Higher Secondary Education — Science Stream 2021 – 2023

Knowledge High School, Nadiad, Gujarat

Achievements & Problem Solving

- Solved **400+ problems on LeetCode** — strong command of algorithms, data structures, and system design under constraint; the same first-principles reasoning FDEs apply to debugging customer environments
- Completed **GfG 160** — 160 consecutive days of structured daily problem solving on GeeksforGeeks

Certifications

- **Advanced Software Engineering Job Simulation** — Forage (Walmart)
- **AWS Solutions Architecture Job Simulation** — Forage
- **DevOps Essentials** — Coursera (IBM)
- **Cybersecurity Fundamentals Specialization** — Coursera (IBM)
- **Machine Learning Essentials** — Let's Upgrade
- **Introduction to Software Development** — Coursera (Amazon)
- **Software Engineering Certificate** — HackerRank